Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Perchonok, Michele Ph.D.		
Project Title:	Handling Procedures of Vegetable Crops		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2002	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

The National Aeronautics and Space Administration (NASA) is working towards future long duration manned space flights beyond low earth orbit. The duration of these missions may be as long as 2.5 years and will likely include a stay on a lunar or planetary surface. The primary goal of the Advanced Food System in these long duration exploratory missions is to provide the crew with a palatable, nutritious, and safe food system while minimizing volume, mass, and waste.

Vegetable crops can provide the crew with added nutrition and variety. These crops do not require any cooking or food processing prior to consumption. The vegetable crops, unlike prepackaged foods, will provide bright colors, textures (crispy), and fresh aromas. Ten vegetable crops have been identified for possible use in long duration missions. They are lettuce, spinach, carrot, tomato, green onion, radish, bell pepper, strawberries, fresh herbs, and cabbage.

Task Description:	Whether these crops are grown on a transit vehicle (e.g., International Space Station) or on the lunar or planetary surface, it will be necessary to determine how to safely handle the vegetables while maintaining acceptability. Recommended handling procedures of all of the proposed vegetable crops have been developed (Toerne and Perchonok, 2002). Since hydrogen peroxide degrades into water and oxygen and is generally recognized as safe (GRAS), hydrogen peroxide has been recommended as the sanitizer. The objective of this research is to determine the required effective concentration of hydrogen peroxide. In addition, it will be determined whether the use of hydrogen peroxide, although a viable sanitizer, adversely affects the quality of the vegetables. Vegetables will be dipped in 1% hydrogen peroxide, 3% hydrogen peroxide, or 5% hydrogen peroxide. Treated produce and controls will be stored in plastic bags at 5oC for up to 14 days. Sensory, color, texture, and total plate count will be measured.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A biodegradable sanitation method for vegetables would provide an alternative to other chemical methodologies.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 01/30/2012)